

3(a)

any **two** from:**I** ref. to two units v one unit **or** bigger v smaller (neutral points)

2

sucrose		fructose
disaccharide R if incorrect monosaccharides stated	v	monosaccharide ;
non-reducing sugar ora A described in terms of reaction with Benedict's reagent	or	reducing sugar ; ora
glycosidic bond present R glucosidic	or	no glycosidic bond present ;
AVP ; e.g. always ring structure(s) hydrolysed / broken down before used as, a respiratory substrate / an energy source	v v	ring structure or straight chain form can be used immediately as, a respiratory substrate / an energy source

Question	Answer	Marks
3(b)(i)	I three fatty acids A fatty acid residue <i>for fatty acid</i> any three from: 1 glycerol ; 2 ester, bond / linkage, qualified ; e.g. labelled on Fig. 3.2 / three ester bonds / ester bond between (O of) glycerol and (C of carboxyl of) fatty acid 3 palmitate is <u>saturated</u> or oleate(s) are, <u>unsaturated</u> / <u>monounsaturated</u> or two fatty acid(s) are, <u>unsaturated</u> / <u>monounsaturated</u>, <u>and</u> one is <u>saturated</u> ; 4 (each fatty acid has a) hydrocarbon chain ; 5 AVP ; extra detail e.g. <i>hydrocarbon chain</i> palmitate 15C <u>and</u> oleate 17C palmitate, $\text{CH}_3(\text{CH}_2)_{14}$ <u>and</u> oleate, $\text{CH}_3(\text{CH}_2)_7\text{CH}=\text{CH}(\text{CH}_2)_7$ palmitate 16C and oleate 18C <i>unsaturated fatty acid tail(s) / oleate</i> <i>ref. to</i>, presence of one, $\text{C}=\text{C}$ / (carbon-carbon) double bond not the maximum number of, hydrogen atoms / hydrogens <i>saturated fatty acid tail(s) / palmitate</i> <i>ref. to</i> no, $\text{C}=\text{C}$ / (carbon-carbon) double bond maximum number of hydrogen atoms / hydrogens	3

Question	Answer	Marks
3(b)(ii)	any one from: <i>idea that</i> provides greater energy per unit volume ; A higher calorific value / greater energy density R produces more energy less mass needed to provide same energy / low(er) mass to energy ratio ; ora higher energy to mass ratio low(er) mass to volume ratio ; AVP ; e.g. glycogen can only be stored in liver and muscles glycogen is hydrated / uses water for storage, so mass increases suggestion that more, areas / adipocytes / adipose tissue, in body to store triglycerides (proportionately) more <u>C-H</u> bonds (to supply H for respiration) <i>suggestion that</i> triglycerides provides insulation during flight (at heights) more compact	1
3(c)	<u>closed, double</u> (circulation / circulatory system) ;	1

Question	Answer	Marks
3(d)	<i>I ref. to metabolism any two from:</i> need more, / large quantities of, oxygen / glucose or more / large quantities, of, oxygen / glucose, delivered / reaches tissues / transported / AW or <i>idea of enough, oxygen / glucose supplied to meet needs ;</i> <i>idea that heart pumps blood (to tissues more) efficiently ;</i> e.g. forcefully / in greater volumes / more / with higher pressure / faster <i>more oxygenated blood is mp2 only</i> suggestion of reason for needing more oxygen or glucose ; I migration e.g. more active fly / wing muscles search for food longer constantly moving / move for longer time generate heat to maintain body temperature, qualified ; e.g. (generally) smaller size / larger SA:V <i>ref. to need to meet demands of carbon dioxide removal (from muscle action / respiration) ;</i> <i>idea that more carbon dioxide produced (so must be removed)</i>	2

Question	Answer	Marks									
3(e)	<i>note: only one mark for both pulmonary vessels</i> <table><tr><th>function of blood vessel</th><th>name of blood vessel</th></tr><tr><td>carries blood from the heart to the lungs</td><td rowspan="2">pulmonary, artery / arteries ; pulmonary, vein / veins</td></tr><tr><td>carries blood to the heart from the lungs</td></tr><tr><td>carries blood from the heart to the rest of body</td><td><u>aorta</u> ; 1 artery</td></tr><tr><td>carries blood to the heart from the rest of body</td><td>1 superior and inferior or anterior and posterior 1 vein venae cavae / vena cava ;</td></tr></table>	function of blood vessel	name of blood vessel	carries blood from the heart to the lungs	pulmonary, artery / arteries ; pulmonary, vein / veins	carries blood to the heart from the lungs	carries blood from the heart to the rest of body	<u>aorta</u> ; 1 artery	carries blood to the heart from the rest of body	1 superior and inferior or anterior and posterior 1 vein venae cavae / vena cava ;	3
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